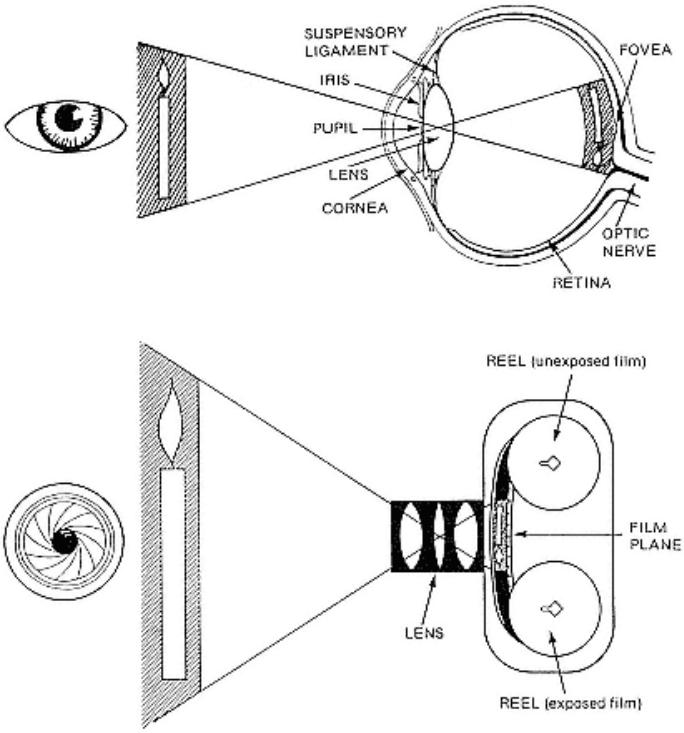
EYE AS A CAMERA SYSTEM:

The human eye can be compared to a camera system, as both the eye and a camera capture the light and convert it into an image.

The main components of the eye that correspond to a camera system include:

1. **The Cornea:** This transparent outer layer of the eye functions like a camera lens, bending light to focus it onto the retina.
2. **The Iris:** The iris functions like the diaphragm in a camera, controlling the amount of light that enters the eye.
3. **The Pupil:** The pupil functions like the aperture in a camera, adjusting the size to control the amount of light entering the eye.
4. **The Retina:** The retina functions like the camera film or sensor, capturing the light and converting it into electrical signals that are sent to the brain.
5. **The Optic Nerve:** The optic nerve functions like the cable connecting the camera to a computer, transmitting the electrical signals from the retina to the brain.



6. **Image Formation:** Both use a **lens** and a **light-sensitive component** to capture light and transform it into an image.

Processing vs. Storage:

1. **Eye:** Processes images for immediate **visual perception**.
2. **Camera:** **Stores** the image for later use.
3. **Complexity:** The eye is significantly more complex with advanced biological functions.
4. **Dynamic Adjustments:** Unlike a basic camera, the eye automatically:
5. Adjusts for varying **light levels**.
6. Changes **focus** dynamically.
7. **Advanced Perception:** The eye perceives **depth and colour**, which provides a richer visual experience.
8. **Real-time Feedback:** The eye adapts to movement and provides a **continuous, real-time image** directly to the brain.

Optical Corrections

Optical corrections refer to devices or techniques used to improve or correct vision problems caused by a refractive error in the eye.

Refractive errors occur when light entering the eye is not properly focused on the retina, leading to blurred vision.

There are several types of refractive errors, including:

1. **Myopia (nearsightedness):** Light is focused in front of the retina, making distant objects appear blurry.
2. **Hyperopia (farsightedness):** Light is focused behind the retina, making near objects appear blurry.
3. **Astigmatism:** Light is not focused evenly on the retina, leading to blurred or distorted vision.

The most common optical corrections include:

1. **Eyeglasses:** Glasses with corrective lenses can be used to refocus light onto the retina, improving vision.
2. **Contact lenses:** Corrective lenses in the form of contacts sit directly on the cornea and work similarly to eyeglasses.
3. **Refractive surgery:** Surgical procedures, such as LASIK and PRK, can reshape the cornea to correct refractive errors.

Optical corrections can greatly improve visual acuity and quality of life for people with refractive errors. However, it is important to have regular eye exams to determine the appropriate correction and monitor eye health.

Lens Materials

The artificial lenses used in cataract surgery or for vision correction can be made of a variety of materials, each with its own unique properties and benefits. The most common lens materials include:

1. **Polymethyl methacrylate (PMMA):** PMMA is a type of plastic that has been used for many years in artificial lenses. It is a durable and affordable material, but does not have the ability to flex and adjust focus like the natural lens.
2. **Silicone:** Silicone is a soft, flexible material that is resistant to cracking and breaking. It is often used in phakic intraocular lenses (IOLs), which are implanted in front of the natural lens.
3. **Acrylic:** Acrylic is a lightweight, clear material that is similar in properties to PMMA. It is often used in foldable IOLs, which can be inserted through a smaller incision.
4. **Hydrophobic acrylic:** Hydrophobic acrylic is a type of acrylic material that has a special surface treatment that helps to reduce glare and halos around lights.
5. **Hydrophilic acrylic:** Hydrophilic acrylic is a type of acrylic material that is designed to be more compatible with the natural fluid in the eye, reducing the risk of vision-threatening complications.

Similarities and differences between human eye and camera:

	Human eye	Camera
Light entry	Pupil	Aperture
Amount of light control	Iris	Diaphragm
Interpretation of image	Retina	Film
Light focused by	Lens	lens

Both the eye and the camera have convex lenses.

1. The eye lens expands or contracts while the camera lens moves forward or backward to focus.
2. The camera lens produces upside down real images while the eye lens produces only upright real images.
3. In both the eye and a camera, the captured light is transformed into an image by the lens and the light-sensitive component.
4. The eye processes the image further, allowing for visual perception, while a camera stores the image for later use.

Eye photoreceptors – Architecture:

1. There are two types of photoreceptors: RODS AND CONES

2. These photoreceptors are localized around an area near the centre of the retina called the macula, which is the functional center of the retina. The fovea is located in the center of the macula.
3. The macula is responsible for sharp, detailed and colour vision, allowing us to see object in front, provided by different types of photoreceptors.

Rod cells	Cone cells
Rod cells are highly sensitive to light and function in night vision (dim light)	Cone cells can detect a wide spectrum of light photons and are responsible for colour vision.
Rods are distributed throughout the retina but there are none at the fovea and blind spot. Rod density is greater in the peripheral retina than in the central retina.	The central fovea is populated exclusively by cones
High amplification	Less amplification
Contains one type of pigment (Rhodopsin)	Contains three types of pigment
Rods cannot distinguish colors	Cones are responsible for colour vision
Rod cells are elongated and cylindrical in shape.	Cone cells are shorter and more conical in shape.
Rod cells have a single long outer segment.	Cone cells have several shorter segments.
Here is estimated to be 92 million rods in retina.	There are 4.6 million cones in the human retina.

BRAIN AS A CPU:

The Nervous System has two main parts:

1. The central nervous system is made up of the brain and spinal cord.
2. The peripheral nervous system is made up of nerves that branch off from the spinal cord and extend to all parts of the body.

Brain is the largest and central organ of the human nervous system and the control unit of the nervous system, which helps us in discovering new things, remembering and understanding, making decisions, and a lot more. The human brain is composed of three major parts.

1. Cerebrum (Forebrain)

The **cerebrum** is the largest part of the brain and is divided into two hemispheres (left and right). It is responsible for high-level functions like **reasoning, emotions, and memory**. It is further divided into four specialized lobes.

a) **Frontal Lobe:** Manages **personality, decision-making, speech** (Broca's area), and voluntary movement.

b) **Parietal Lobe:** Interprets sensory information like **touch, temperature, and pain**, and helps with spatial orientation.

c) **Temporal Lobe:** Essential for **hearing, language comprehension** (Wernicke's area), and memory storage.

d) **Occipital Lobe:** Dedicated to **visual processing**, including the recognition of shapes and colours.

2. Cerebellum (Hindbrain)

Located at the back of the head, the **cerebellum** is often called the "little brain". Its primary functions include.

a) **Coordination:** Fine-tunes voluntary muscle movements for precision.

b) **Balance and Posture:** Maintains the body's equilibrium and steady posture

3. Brainstem

The **brainstem** connects the cerebrum and cerebellum to the spinal cord. It acts as a relay station and controls **involuntary, life-sustaining functions**.

a) **Midbrain:** Coordinates eye movements and processes basic visual and auditory reflexes.

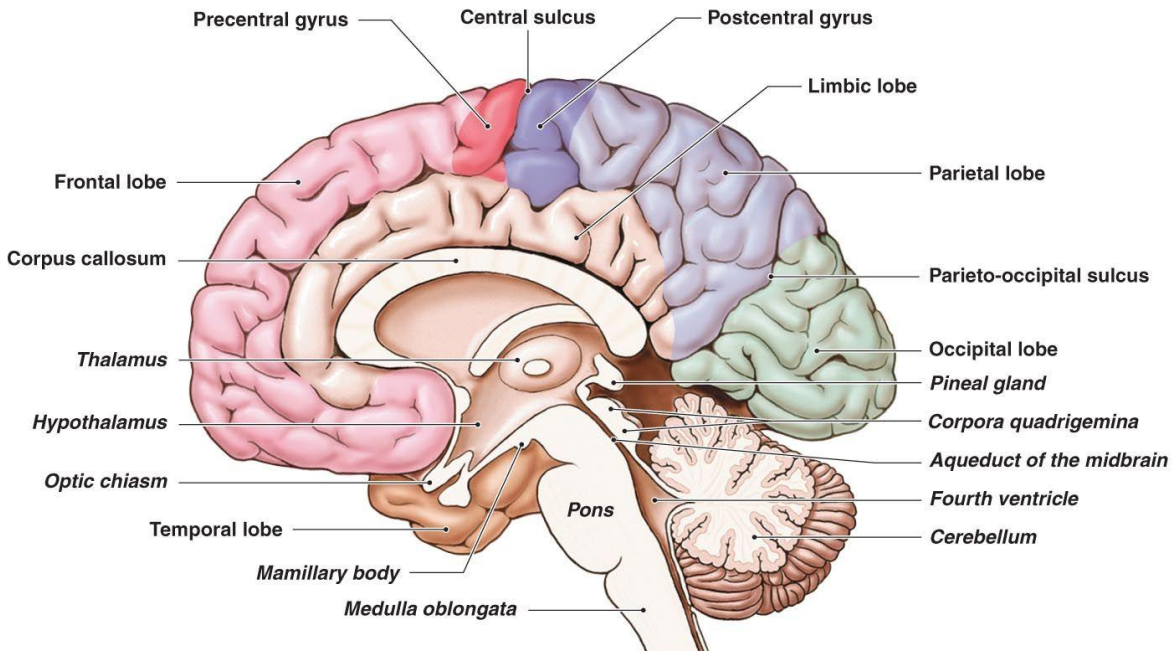
b) **Pons:** Acts as a bridge for signal transmission and helps regulate **breathing**.

c) **Medulla Oblongata:** Controls vital autonomic processes such as **heart rate, blood pressure, and swallowing**.

Other Key Internal Structures

1) **Thalamus:** Acts as a relay station for almost all sensory information going to the cerebrum.
2) **Hypothalamus:** Regulates core body functions like **hunger, thirst, sleep**, and body temperature.
3) **Pituitary Gland:** Known as the "master gland," it produces hormones that control growth and other glands.
4) **Spinal Cord**
   a) The spinal cord is a cylindrical bundle of nerve fibers and associated tissues enclosed within the spine and connect all parts of the body to the brain.
   b) It begins in continuation with the medulla and extends downwards. It is enclosed in a bony cage called vertebral column and surrounded by membranes called meninges.
   c) The spinal cord is concerned with spinal reflex actions and the conduction of nerve impulses to and from the brain.

A midsagittal view showing the inner boundaries of the lobes of the cerebral cortex
(Structures outside of the cerebrum are labeled in italics.)



Brain as CPU

The brain's architecture can be mapped to a computer's functional units:

1) **Control Unit (CU) →Thalamus & Hypothalamus**: These relay sensory and motor signals to the correct areas, coordinating bodily functions similar to a CU managing instruction flow.
2) **Arithmetic Logic Unit (ALU) → Cerebral Cortex**: High-level information processing, logical reasoning, and decision-making occur here.
3) **Memory Unit → Hippocampus & Synapses**: The Hippocampus manages short-term "cache" memory, while long-term data is stored in the strength of synaptic connections throughout the cortex.
4) **Input/Output (I/O) → Nervous System**:
    - **Input**: Sensory organs (eyes, ears, etc.) provide data.
    - **Output**: Muscles and glands execute the processed commands

Comparison:

Basis for Comparison	Brain	Computer
Construction	Neurons and synapses	ICs, transistors, diodes, capacitors, transistors, etc.

Memory growth	Increases each time connecting synaptic links	Increases by adding more memory chips
Backup systems	Built-in backup system	Backup system is constructed manually
Memory power	—	100 million megabytes
Memory density	7	10^{14} bits/cm³
Energy consumption	12 watts of power	Gigawatts of power
Information storage	Stored in electrochemical and electric impulses.	Stored in numeric and symbolic form (i.e. in binary bits).
Size and weight	The brain's volume is 1500 cm³ and weight is around 3.3 pounds.	Variable weight and size from few grams to tons.
Information processing	Uses chemicals to fire the action potential in the neurons.	Communication is achieved through electrical coded signals.
Power	Low	High
Input/output equipment	Sensory organs	Keyboards, mouse, web cameras, etc.
Structural organization	Self-organized	Pre-programmed structure
Parallelism	Massive	Limited
Reliability and damageability	Brain is self-organizing, self-maintaining and reliable.	Computers perform a monotonous job and can't correct itself.

Peripheral Nervous System

Peripheral nervous system involves the parts of the nervous system outside the brain and the spinal cord. It develops from the central nervous system which connects different parts of the body with the CNS. It carries out both voluntary and involuntary actions with the help of peripheral nerves.

PNS includes two types of nerve fibres:

1. Afferent nerve fibres – These are responsible for transmitting messages from tissues and organs to the CNS.

2. Efferent nerve-fibres – These are responsible for conveying messages from CNS to the corresponding peripheral organ.

a. **Somatic neural system (SNS):** It is the neural system that controls the voluntary actions in the body by transmitting impulses from CNS to skeletal muscle cells. It consists of the somatic nerves.

b. **Autonomic Nervous System (ANS):** The autonomic nervous system relays impulses from the central nervous system to the involuntary organs and smooth muscles of the body (involuntary actions like digestion, respiration, salivation, etc.).

HEART AS A PUMP SYSTEM:

Heart is sort of like a pump, or two pumps in one. The right side of your heart receives blood from the body and pumps it to the lungs. The left side of the heart does the exact opposite: It receives blood from the lungs and pumps it out to the body.

The parts of your heart are like the parts of a building.

1. **Heart Walls:** Your heart walls are the muscles that contract (squeeze) and relax to send blood throughout your body
2. **Heart chambers:** Your heart has four separate chambers. You have two chambers on the top (atrium, plural atria) and two on the bottom (ventricles), one on each side of your heart.
3. **Heart valves:** Your heart valves are like doors between your heart chambers. They open and close to allow blood to flow through.
4. **Blood vessels:** Your heart pumps blood through three types of blood vessel.
5. **Vena Cava:** The vena cava is divided into two primary sections based on the region of the body they drain:

   a) **Superior Vena Cava (SVC):** This large, short vein carries blood from the **head, neck, arms, and upper chest**. It enters the top of the right atrium.

   b) **Inferior Vena Cava (IVC):** The **largest vein in the body**, it carries blood from the **legs, feet, and abdominal/pelvic organs**. It is much longer and wider than the SVC and enters the bottom of the right atrium.

6. **Electrical conduction system:** Your heart's conduction system is like the electrical wiring of a building. It controls the rhythm and pace of your heartbeat. Signals start at the top of your heart and move down to the bottom**.**
7. **Circulation of blood:** Blood is pumped through these chambers and valves in the following way.

   a) Blood returning to the heart from the body (by way of the vena cava) flows into the right atrium.

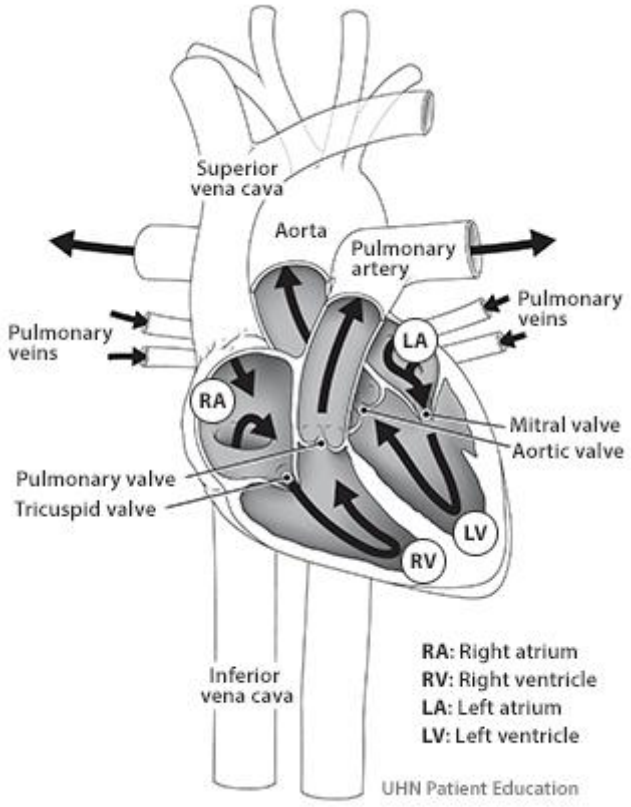
   b) When the atrium contracts, it forces blood through the tricuspid valve into the right ventricle.

   c) From the right ventricle, it is pumped through the pulmonary artery to the lungs where it picks up oxygen.

   d) From the lungs, it returns to the heart through the left atrium.

e) From the left atrium, it goes through the mitral valve to the left ventricle.

f) The left ventricle pumps the blood into the aorta and then to all parts of the body.



Common heart related issues:

1. **Arrhythmias:** Abnormalities in the heart's rhythm or rate can be detected using an ECG.
2. **Heart disease:** Changes in the heart's electrical activity can indicate the presence of heart disease, such as coronary artery disease or heart attacks.
3. **Heart attack:** An ECG can help diagnose a heart attack by detecting changes in the heart's electrical activity that indicate a lack of blood flow to the heart.
4. **Angina** is a type of chest pain caused by reduced blood flow to the heart. Angina is a symptom of coronary artery disease.
5. **Atherosclerosis** is the buildup of fats, cholesterol and other substances in and on the artery walls. This buildup is called plaque.
6. **Heart failure:** Condition in which heart is unable to supply oxygenated blood throughout the body.
7. **Coronary artery disease**: If the blood supplying arteries become narrow or blocked due to plaque leads to coronary artery diseases such as high blood pressure, high cholesterol.
8. **Valve disorders:** When the valves in the heart do not work properly.

Reasons for Blockages of Blood Vessels:

a) **High cholesterol levels:** Excessive amounts of low-density lipoprotein (LDL) cholesterol in the blood can lead to the formation of plaque in the blood vessels, which can narrow or block them.

b) **High blood pressure:** Over time, high blood pressure can cause damage to the blood vessels, leading to the formation of plaque and blockages.

c) **Smoking:** Smoking can damage the inner walls of blood vessels and promote the buildup of plaque, leading to blockages.

d) **Diabetes:** People with uncontrolled diabetes are at a higher risk of developing blockages in their blood vessels, due to damage to the blood vessels from high levels of glucose.

e) **Age:** As people age, the blood vessels can become stiff and less flexible, increasing the risk of blockages.

f) **Genetics:** Some people may be predisposed to developing blockages in their blood vessels due to genetic factors.

g) **Poor diet:** A diet high in saturated fats, trans fats, and cholesterol can increase the risk of developing blockages in the blood vessels.

Stents:

1. Stents are small, metal mesh devices that are used to treat blockages in blood vessels.
2. They are typically used in procedures such as angioplasty, where a balloon catheter is used to open up a blocked blood vessel and a stent is placed to keep it open.
3. The design of stents can vary depending on the specific medical condition it is used to treat.
4. Some are made out of fabric. These are called stent grafts and are often used for larger arteries.
5. Others are made of a material with drug-eluting stent that dissolves, and your body absorbs it over time.

Design of stents

1. **Shape:** Stents can be designed in a variety of shapes, including cylindrical, helical and spiraled, to match the shape of the blood vessel and provide adequate support.
2. **Material:** Stents can be made of different materials, including stainless steel, cobalt chromium, and Nitinol (a type of metal that is flexible and can return to its original shape after being expanded).
   - The selection of material depends on the factors such as biocompatibility, durability, ability to deliver drug & other therapeutic agents.
3. **Coating:** Stents can be coated with different materials or with drugs to prevent the formation of blood clots and reduce the risk of re narrowing or promote healing.
4. **Delivery system**: Delivered via catheter- based system.

5. **Expansion mechanism**: Stents can be designed to expand in different ways, such as by balloon inflation or self-expansion, depending on the type of stent and the specific medical condition it is used to treat.

Natural pacemaker

1. An electrical stimulus is generated in a special part of the heart muscle called the sinoatrial node (SA node).
2. The sinus node is a small mass of special tissue in the right upper chamber of the heart (right atrium) which is natural pacemaker of the heart.
3. Then the electric impulses travel to the atrioventricular node (AV node) present between atria & ventricles, allow then to pass through ventricles.
4. Then these impulses are carried by Purkinje fibre, which coordinate the contraction of ventricles & helps in blood pumping.
5. In an adult, the sinus node sends out a regular electrical pulse 60 to 100 times per minute.

Pace Makers

1. A pacemaker is a small device that is surgically implanted in the chest that generate electrical impulses to regulate the heartbeat.
2. It is used to treat heart rhythm disorders, such as bradycardia (a slow heartbeat) or tachycardia (fast heartbeat), by delivering electrical impulses. to the heart to regulate its rhythm.

Design:

1. **Pulse generator:** This small metal container houses a battery and the electrical circuitry that controls the rate of electrical pulses sent to the heart.
2. **Leads (electrodes):** One to three flexible, insulated wires are each placed in one or more chambers of the heart and deliver the electrical pulses to adjust the heart rate. However, some newer pacemakers don't require leads. These devices, called leadless pacemakers, are implanted directly into the heart muscle.

Types of pace maker

1. **Single chamber pacemaker:** This type usually carries electrical impulses to the right ventricle of your heart.
2. **Dual chamber pacemaker:** This type carries electrical impulses to the right ventricle and the right atrium of your heart to help control the timing of contractions between the two chambers.

3. **Bi-ventricular pacemaker:** Biventricular pacing is for people who have heart failure and heartbeat problems. This type of pacemaker stimulates both of the lower heart chambers (the right and left ventricles) to make the heart beat more efficiently.

Electrocardiogram (ECG)

1. An electrocardiogram is a painless, non-invasive way to help diagnose many common heart problems.
2. The heart's pumping action is controlled by electrical signaling, which generates the rhythm of the heartbeat.
3. This electrical signaling can be monitored using an electrocardiogram (ECG), which records the electrical activity of the heart with electrodes placed on the chest region.
4. An ECG measures the electrical signal and generates a trace or waveform that reflects the electrical activity of the heart.
5. This trace/waveform can be used to diagnose heart conditions and monitor the heart's function.

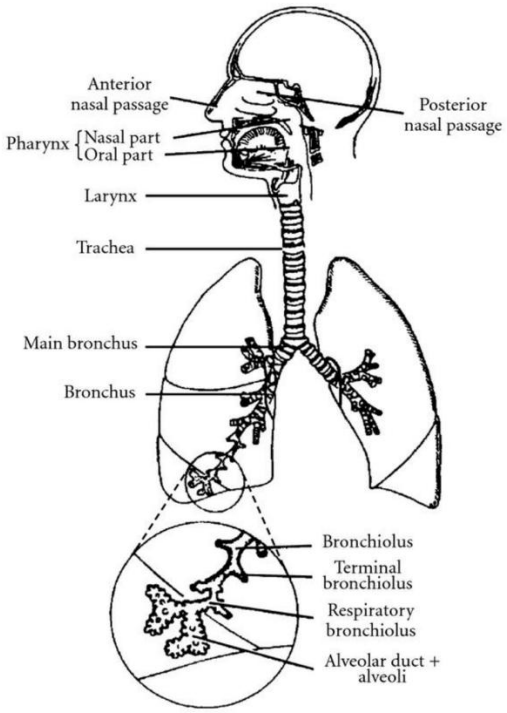
LUNG AS PURIFICATION SYSTEM:

The lung purifies air by removing harmful substances and adding oxygen to the Bloodstream

1. **Filtration:** The nose and mouth serve as a first line of defense against harmful substances in the air, such as dust, dirt, and bacteria.
2. **Moisturization:** The air is also humidified as it passes over the moist lining of the respiratory tract, which helps to keep the airways moist and prevent them from drying out.
3. **Gas Exchange**: Once the air reaches the alveoli, the gas exchange process occurs where oxygen diffuses across the thin alveolar and capillary walls into the bloodstream, and carbon dioxide diffuses in the opposite direction, from the bloodstream into the alveoli to be exhaled.

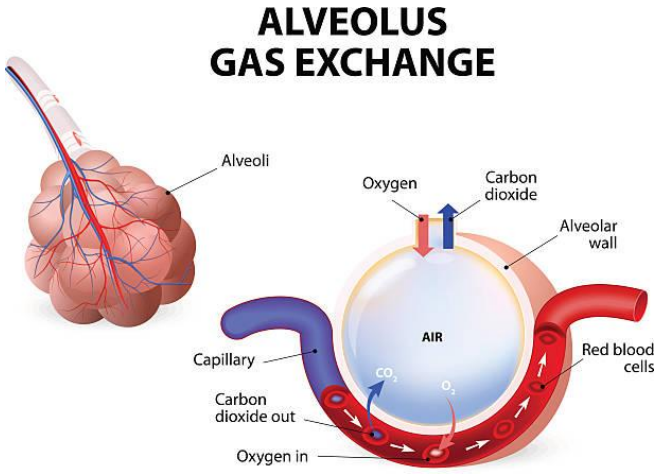
Architecture of lungs

Air enters the body through the mouth or nose (nasal cavity) and quickly moves to the pharynx. From there, it passes through the larynx (voice box) and enters the trachea. Within the lungs, the trachea branches into a left and right bronchus. These further divide into smaller and smaller branches called bronchioles. The smallest bronchioles end in tiny air sacs. These are called alveoli.



1. **Trachea**: The trachea is the main airway that leads from the larynx (voice box) to the lungs. It is lined with cilia and mucus-secreting glands that help to filter out harmful substances and trap them in the mucus. The trachea is a strong tube that contains rings of cartilage that prevent it from collapsing.
2. **Bronchi:** The trachea branches into two main bronchi, one for each lung. The bronchi are larger airways that continue to branch into smaller airways called bronchioles.
3. **Bronchioles**: The bronchioles are smaller airways that eventually lead to the alveoli. They are surrounded by tiny air sacs called alveoli, which are the sites of gas exchange.
4. **Alveoli**: The alveoli are tiny air sacs that are lined with a network of capillaries. This close proximity of the alveoli and capillaries allows for efficient diffusion of oxygen and carbon dioxide between the air in the alveoli and the bloodstream.
5. **Gas exchange mechanism in lungs:**
   a. The gas exchange mechanism in the lung involves the transfer of oxygen from the air in the alveoli to the bloodstream, and the transfer of carbon dioxide from the bloodstream to the air in the alveoli.
   b. This process happens by diffusion and occurs due to differences in partial pressures of oxygen and carbon dioxide.
6. **Oxygen Diffusion:** The concentration of oxygen in the air in the alveoli is higher than the concentration of oxygen in the bloodstream. This difference creates a gradient that causes oxygen to diffuse from the alveoli to the blood in the capillaries where it binds to hemoglobin in red blood cells to form oxyhemoglobin.

7. Carbon Dioxide Diffusion: The concentration of carbon dioxide in the bloodstream is higher than the concentration of carbon dioxide in the air in the alveoli. This difference creates a gradient that causes carbon dioxide to diffuse from the bloodstream into the alveoli, where it is exhaled.



Oxygen and Carbon diffusion in alveoli

Spirometry

a. Spirometry is a diagnostic test that measures the function of the lungs by measuring the amount and flow rate of air that can be exhaled.
b. The test is commonly used to diagnose lung conditions such as asthma, chronic obstructive pulmonary disease (COPD), and interstitial lung disease.

Principle:

1. The principle behind spirometry is to measure the volume of air that can be exhaled from the lungs in a given time period.
2. By measuring the volume of air exhaled. spirometry can provide information about the functioning of the lungs and the ability of the lungs to move air in and out.

Working:

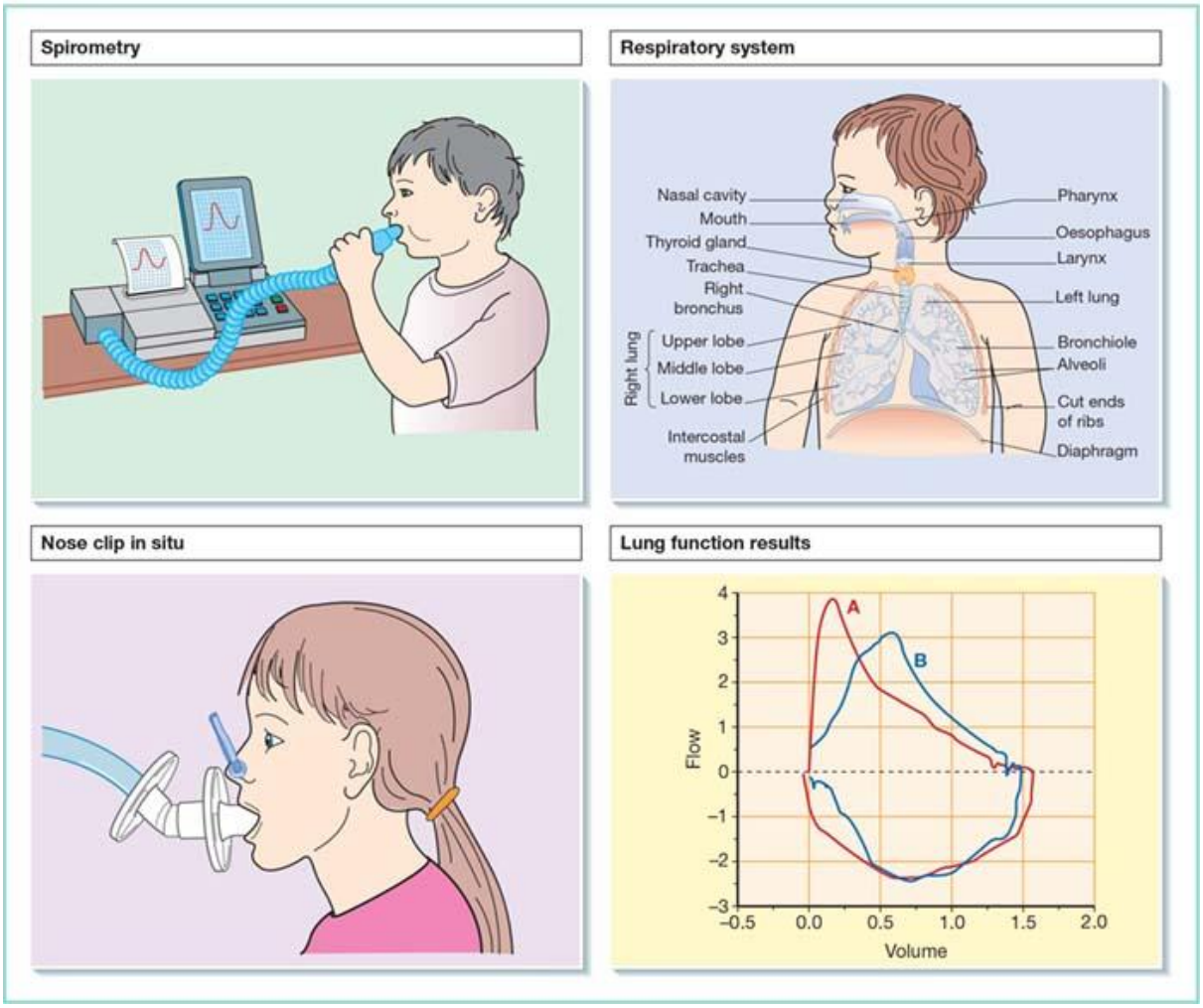
1. Spirometry is performed using a spirometer, a device that consists of a mouthpiece, a flow sensor, and a volume sensor.
2. The patient is asked to exhale as much air as possible into the spirometer after a deep breath, and the spirometer measures the volume and flow rate of the exhaled air.
3. The volume of air exhaled is displayed on a graph called a flow-volume loop, which provides information about the lung function.

Spirometry measures two main components:

a. **Forced vital capacity (FVC):** FVC is the highest amount of air you can breathe out after taking a deep breath in.

b. **Forced expiratory volume (FEV1):** FEV1 is the amount of air you breathe out in one second.

c. The FEV1/FVC ratio parameter is calculated and is converted to percentage which tells us about the lung functionality.



Abnormal Lung Physiology

a. Abnormal lung physiology refers to any deviation from the normal functioning of the respiratory system.

b. This can be caused by a variety of factors including diseases, injuries or genetic conditions. Some common examples of abnormal lung physiology include

i. **Asthma:** A chronic inflammatory disease that causes the airways to narrow, making it difficult to breathe.

ii. **Chronic obstructive pulmonary disease (COPD):** A progressive lung disease that makes it hard to breathe and can include conditions such as emphysema and chronic bronchitis.

iii. **Pulmonary fibrosis:** A disease in which scar tissue builds up in the lungs, making it difficult to breathe and reducing lung function.

iv. **Pneumonia:** An infection in the lungs that can cause inflammation and fluid buildup in the air sacs.

v. **Pulmonary embolism:** A blockage in one of the pulmonary arteries, usually by a blood clot, which can cause lung damage and reduce oxygen flow to the body.

vi. **Lung cancer:** A type of cancer that originates in the lung and can impair lung function by interfering with normal air flow and oxygen exchange.

Chronic obstructive pulmonary disease (COPD):

1. A progressive lung disease that makes it hard to breathe and can include conditions such as emphysema and chronic bronchitis.
2. In COPD, the airways and small air sacs (alveoli) in the lungs become damaged or blocked, leading to difficulty in exhaling air. This results in a decrease in lung function, leading to shortness of breath, wheezing, and coughing.
3. The primary cause of COPD is long-term exposure to irritants such as tobacco smoke, air pollution and dust, a history of frequent lung infections, a family history of lung disease, and exposure to second-hand smoke.
4. There is no cure for COPD
5. Treatment options include use of medications such as bronchodilators and steroids, oxygen therapy, and lung rehabilitation. In severe cases, surgery may also be an option. In addition, quitting smoking and avoiding exposure to irritants is crucial in managing COPD.

Ventilator

1. They are commonly used in the treatment of acute respiratory failure, which can occur as a result of a variety of conditions such as pneumonia, severe asthma, and chronic obstructive pulmonary disease (COPD)
2. Ventilators work by delivering pressurized air or oxygen into the lungs through a breathing tube or mask.
3. The pressure can be adjusted to match the patient's needs and to help maintain adequate oxygen levels in the blood.
4. While ventilators can be lifesaving for individuals with acute respiratory failure, they also come with potential risks and complications.

Heart lung machine

1. A heart-lung machine is a piece of equipment that temporarily takes over the work of the heart and/or lungs, providing blood and oxygen to the body.
2. It is most often used during serious procedures that require the heart to be stopped.
3. Patients are kept on a heart-lung machine for only as long as it takes to stop the heart from beating, complete open-heart surgery or a procedure on the lungs, and restart the heart.

Working:

1. **Blood circulation**: It circulates the patients' blood through a series of tubing & an oxygenator, which serves as a temporary replacement for the lungs.
2. **Temperature control:** it controls the patients' blood temperature.
3. **Blood pressure control:** Machine is programmed to control the flow rate & pressure of blood.
4. **Gas exchange**: Oxygenator in machine temporarily serves as patients' lungs.
5. **Monitoring:** It includes monitors & sensors to measure BP, Oxygen level, heart rate.

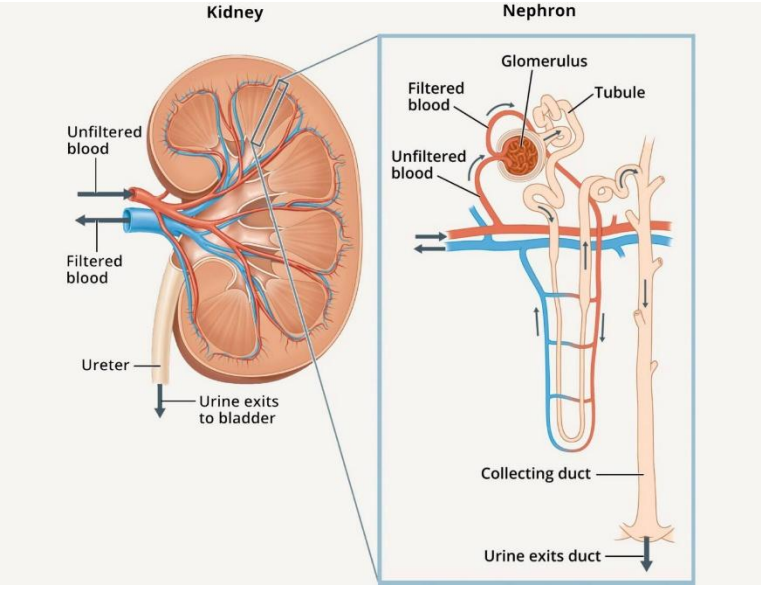
KIDNEY AS A FILTRATION SYSTEM:

Functions of kidneys

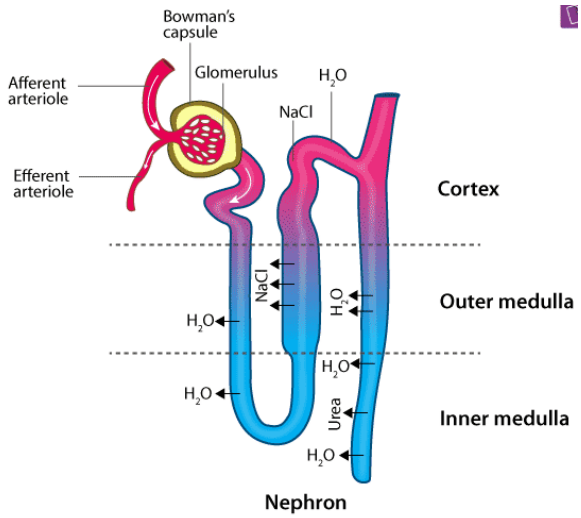
1. It removes waste and excess fluid from the bloodstream and maintains a delicate balance of electrolytes, hormones, and other substances that are critical for the body's normal functioning.
2. Kidneys also make hormones that help control blood pressure, make red blood cells, keeps bones strong and healthy.

Architecture of kidneys

1. The kidneys are two bean-shaped organs, each about the size of a fist.
2. They are located just below the rib cage, one on each side of the spine.
3. The kidney has three regions- renal cortex, renal medulla and renal pelvis.
4. Renal cortex is the external part of the kidney and encloses nephrons, renal medulla is the inner part, consists of pyramid-like structures which encloses the collecting tubules of nephrons, renal pelvis is a funnel shaped structure which sends urine to ureters.
5. The urine flows from the kidneys to the bladder through two thin tubes of muscle called ureters, one on each side of the bladder.
6. Urinary bladder stores urine and is expelled out through urethra when there is an urge to pass urine.



Structure of nephron/ Mechanism of filtration:



Nephrons: Functional unit of kidney, responsible for filtering the blood. Each nephron includes a filter, called the glomerulus, and a tubule.

a. **The nephrons work through a two-step process:** The glomerulus filters blood, and the tubule returns needed substances to your blood and removes wastes.
b. **Bowman's capsule:** This is a cup-shaped structure that surrounds the glomerulus and filters waste and excess fluid from the bloodstream into the renal tubule.
c. **Glomerulus:** A network of tiny blood vessels within the Bowman's capsule that filters waste and excess fluid from the bloodstream.
d. **Loop of Henle:** A U-shaped segment of the renal tubule that is critical for the reabsorption of ions and water.
e. **Proximal convoluted tubule:** A segment of the renal tubule that reabsorbs important substances such as glucose, amino acids, and electrolytes, back into the bloodstream.

f. **Distal convoluted tubule:** A segment of the renal tubule that regulates the levels of electrolytes and other important substances in the bloodstream.

g. **Collecting duct:** A series of ducts that collect the filtrate from the renal tubules and transport it to the renal pelvis, where it drains into the ureter and eventually into the bladder.

h. The nephrons are surrounded by a network of blood vessels, including the afferent arteriole and the efferent arteriole, which bring blood into and out of the glomerulus, respectively. The filtrate produced by the nephron passes through the renal tubules, where it is modified by reabsorption and secretion, before being eliminated from the body as urine.

Chronic Kidney Disease (CKD)

1. Chronic kidney disease includes conditions that damage your kidneys with decreased ability to filter wastes from the blood. wastes build up to high levels in your blood and make you feel sick.
2. Caused by a variety of factors, including diabetes, high blood pressure, and other health problems that damage the kidneys.
3. Other complications are: high blood pressure, anemia (low blood count), weak bones, poor nutritional health and nerve damage
4. As the disease progresses, it can lead to more serious complications, such as anemia, nerve damage, and an increased risk of heart disease and stroke.
5. **Treatment -** lifestyle changes - eating a healthy diet and exercising regularly, as well as medications to manage symptoms and underlying health conditions.
6. In severe cases, kidney transplant or dialysis may be necessary.

Dialysis System:

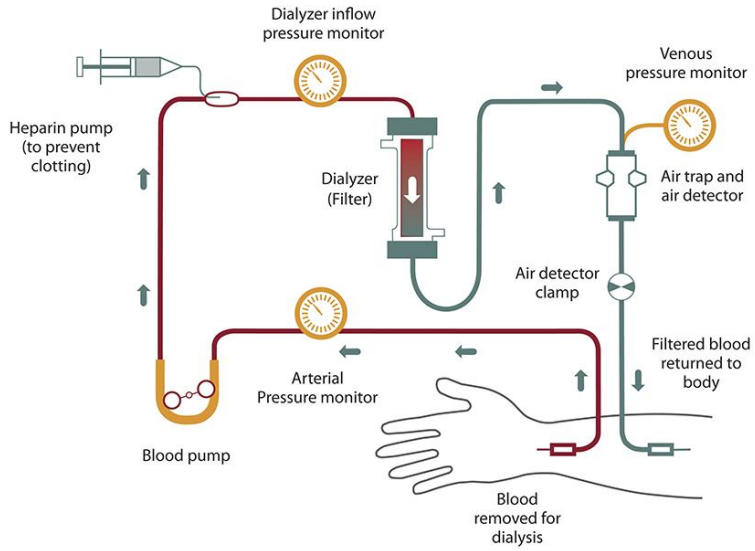
Dialysis is a medical treatment that helps to filter waste and excess fluids from the blood when the kidneys are unable to function properly.

There are two main types of dialysis systems:

1. **Hemodialysis** is a treatment to filter wastes and water from your blood.
   a. During hemodialysis, the blood is drawn out of the body and is passed through a heparin pump (anticoagulant).
   b. It is then passed through a dialysis machine which contains a semi permeable membrane made up of cellophane.
   c. The membrane is surrounded by dialysis fluid which is having the same concentration of blood plasma.
   d. Hence allows waste products and excess fluids to pass through it to the dialyzing fluid retaining proteins and blood cells in the blood.

e. The filtered blood is then returned to the body.

f. Hemodialysis typically takes place in a hospital or dialysis center, and is typically performed three times a week for three to four hours at a time.



Schematic structure of hemodialysis

2. Peritoneal dialysis is a type of dialysis that uses the peritoneum in a person's abdomen as the membrane through which fluid and dissolved substances are exchanged with the blood.

a. The process involves pumping dialysis fluid (dialysate) into the space inside your abdomen (peritoneum) through a catheter.

b. The dialysate remains in the abdomen for several hours and absorbs waste products, excess fluid from the patient's blood. The used dialysate is then drained out from the abdomen.

c. A sterile solution is introduced into the abdomen, where it absorbs waste and excess fluids, and is then drained and replaced with fresh solution.

d. Peritoneal dialysis can be performed at home and allows for more flexibility in scheduling.

e. Both hemodialysis and peritoneal dialysis can effectively treat the symptoms of kidney failure, but each has its own advantages and disadvantages.

